

PACKAGED HEAT PUMP

ZHD

RAIDER® ROOFTOP UNITS

Standard Efficiency | **R-454B** | 60Hz

COMMERCIAL

PRODUCT SPECIFICATIONS (EHB)



7.5 to 10 Tons

Net Cooling Capacity - 87,000 to 118,000 Btuh

Net Heating Capacity - 83,000 to 116,000 Btuh

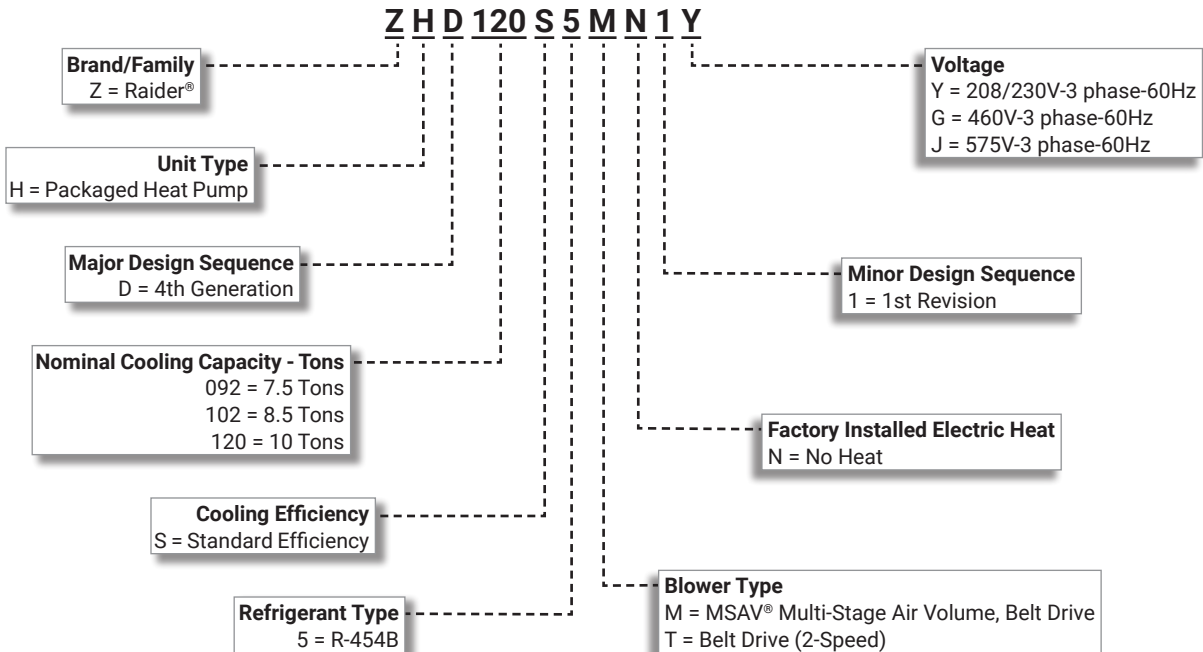
Optional Electric Heat - 7.5 to 60 kW

RAIDER



ASHRAE
Standard
90.1

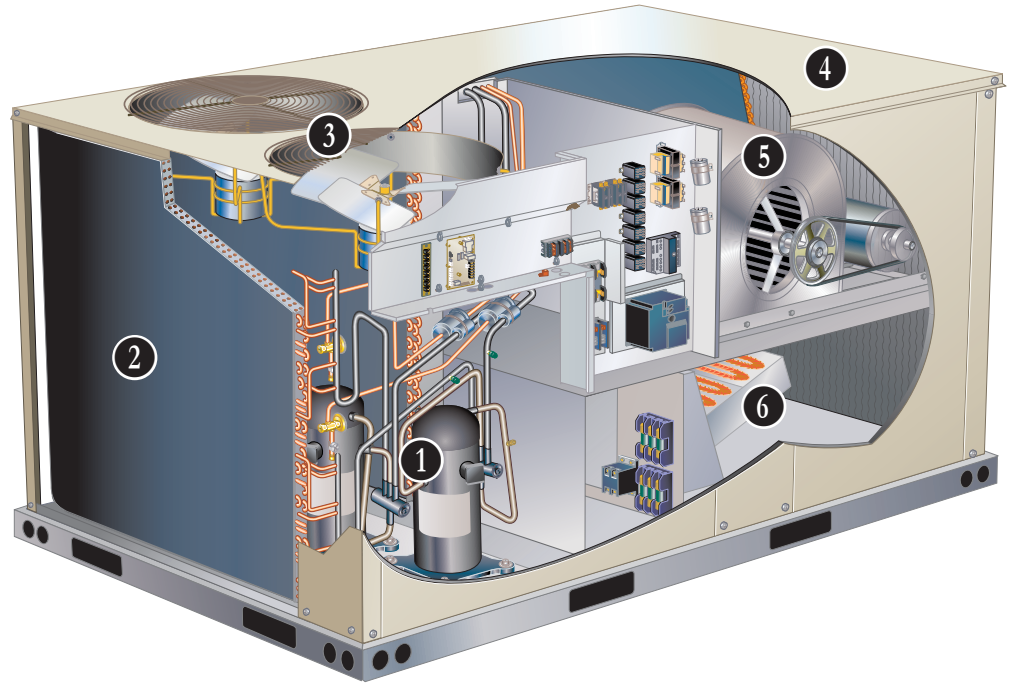
MODEL NUMBER IDENTIFICATION



FEATURE HIGHLIGHTS

Raider® rooftop units from Lennox Commercial are the new standard for cost efficient reliable, efficient rooftop units built for long-lasting performance that can significantly improve indoor environments.

1. Scroll Compressors
2. Copper Tube Coil Construction
3. Outdoor Coil Fans
4. Heavy Gauge Steel Cabinet
5. MSAV® Multi-Stage Air Volume Blower
6. Electric Heat (option)



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APPROVALS AND WARRANTY

APPROVALS

- AHRI Standard 340/360-2023 certified
- ETL Intertek listed
- Unit and components are ETL, NEC, and CEC bonded for grounding to meet safety standards for servicing
- All models are ASHRAE 90.1 energy efficiency compliant and meet or exceed requirements of Section 6.8
- All models meet DOE 2023 energy efficiency standards and UL 60335-2-40 Refrigerant Detector Requirements
- All models meet California Code of Regulations, Title 24 and ASHRAE 90.1 Section 6.4.3.10 requirements for staged airflow
- ISO 9001 Registered Manufacturing Quality System

WARRANTY

- Compressors - Limited five years
- Variable-Frequency Drive (VFD) (optional) - Limited five years
- High Performance Economizers (optional) - Limited five years
- All other covered components - Limited one year

FEATURES AND BENEFITS

COOLING/HEATING SYSTEM

- Designed to maximize sensible and latent cooling performance at design conditions
- System can operate from 30°F to 125°F without any additional controls

R-454B Refrigerant

- Low GWP (Global Warming Potential)
- Zero ODP (Ozone Depletion Potential)
- Low Toxicity/Lower Flammability - A2L
- Unit is factory pre-charged

1 Scroll Compressors

- System consists of one two-stage scroll compressor and one single-stage scroll compressor
- Resiliently mounted on rubber grommets for quiet operation

Thermal Expansion Valves

- Ensures optimal performance throughout the application range
- Removeable power element

Filter/Driers

- High capacity filter/drier protects the system from dirt and moisture

Reversing Valve

- 4-way interchange reversing valve rapidly changes the direction of refrigerant flow resulting in quick changeover from cooling to heating and vice versa

Defrost Control

- Provides a defrost cycle, if needed, every 30 or 60 or 90 minutes (adjustable) of compressor "on" time at outdoor coil temperature below 35°F
- Pressure and temperature switches terminate defrost cycle

High Pressure Switches

- Protects the compressor from overload conditions such as dirty outdoor coils, blocked refrigerant flow or loss of outdoor fan operation

Low Pressure Switches

- Protects the compressor from conditions such as low refrigerant charge, or loss of indoor air flow.

Coil Construction

- 2** • Copper tube construction
- Enhanced rippled-edge aluminum fins
- Flared shoulder tubing connections
- Silver soldered construction
- Factory leak tested

Indoor Coil

- Cross-row circuiting with rifled tubing optimizes both sensible and latent cooling capacity

Antimicrobial Condensate Drain Pan

- Composite pan, sloped to meet drainage requirements of ASHRAE 62.1
- Antimicrobial additive resists growth of mold and mildew on drain pan, which improves indoor air quality and reduces drain line blockage
- Side drain connections

Outdoor Coil Fan Motors

- Thermal overload protected
- Totally enclosed
- Permanently lubricated ball bearings
- Shaft up
- Wire basket mount

3 Outdoor Coil Fans

- PVC coated fan guard furnished

FEATURES AND BENEFITS

COOLING/HEATING SYSTEM (continued)

Required Selections

Cooling Capacity

- Specify nominal cooling capacity

Options/Accessories

Field Installed

Condensate Drain Trap

- Available in copper or PVC

Drain Pan Overflow Switch

- Monitors condensate level in drain pan
- Shuts down unit if drain becomes clogged

Low Ambient Kit (0°F)

- Cycles the outdoor fan while allowing compressor operation in the cooling cycle
- Intermittent fan operation allows the system to operate without icing the indoor coil and losing capacity
- Designed for use in ambient temperatures no lower than 0°F

LOW GWP REFRIGERANT DETECTION SYSTEM (RDS)

- Complies with UL 60335-2-40 approved standard
- Required for all systems using R-454B refrigerant
- Factory installed on all units
- Consists of a refrigerant detection sensor(s) and a mitigation control
- Ensures safe operation for systems equipped with R-454B refrigerant
- Sensor(s) monitors indoor coil area for R-454B refrigerant
- If R-454B refrigerant is detected the refrigerant detection system will prevent compressor and heating operation until R-454B refrigerant is no longer detected
- Refrigeration detection system energizes blower if any R-454B refrigerant is detected to mitigate any concentrations of refrigerant from the unit and the system

CABINET

Construction

- 4 • Heavy-gauge steel panels
- Full perimeter heavy-gauge galvanized steel base rail
- Base rails have rigging holes
- Three sides of the base rail have forklift slots
- Raised edges around duct and power entry openings in the bottom of the unit for water protection

Airflow Choice

- Units are shipped in downflow (vertical) return air flow configuration

Duct Flanges

- Provided for horizontal duct attachment.

Power Entry

- Electrical lines can be routed through the unit base or through horizontal access knock-outs

NOTE - Optional Bottom Power Entry Kit is available.

Exterior Panels

- Constructed of heavy-gauge, galvanized steel
- Textured pre-paint with polyurethane finish
- Cyclic salt fog and UV exposure up to 1,680 hours per ASTM D5894

Insulation

- Fully insulated with non-hygroscopic fiberglass insulation (conditioned areas)

Access Panels

- Filter section
- Blower/heating section
- Compressor/controls section
- Recessed handles for easy service access

Options/Accessories

Factory Installed

Corrosion Protection

- Completely flexible immersed coating
- Electrodeposited dry film process (AST ElectroFin E-Coat)
- ASTM B117 / DIN 53167 Salt Spray - 15,000+ hours
- ASTM G85 Annex A3 SWAAT Modified Salt Spray - 3000 hours
- VA Master Construction Specification Division 23 for High Humidity Installations
- CID AA-52474A (GSA)
- Indoor Corrosion Protection:
 - Coated coil
- Outdoor Corrosion Protection:
 - Coated coil

FEATURES AND BENEFITS

CABINET (continued)

Field Installed

Combination Coil/Hail Guards

- Heavy gauge steel frame
- Painted to match cabinet
- Expanded metal mesh protects outdoor coil

BLOWER

A wide selection of supply air blower options are available to meet a variety of airflow requirements

5 Blower Motor

- Overload protected
- Ball bearings
- Available in several different sizes to maximize air performance.
- Two-speed belt drive motor (low static/high static) is available on 092-102 models

Supply Air Blower

- Forward curved blades
- Double inlet
- Blower wheel statically and dynamically balanced
- Ball bearings
- Adjustable pulley (allows speed change).
- Blower assembly slides out of unit for servicing

MSAV® Multi-Stage Air Volume Operation (092 and 102 Models)

- MSAV® Multi-Stage Air Volume stages the amount of airflow according to compressor stages, heating demand and ventilation demand
- Units utilize a 2-Speed induction blower motor to stage the supply air blower airflow
 - Low Speed - 1st and 2nd Stage cooling and ventilation mode
 - High Speed - 3rd Stage (Full load) cooling and all heat modes
- High speed blower operation is set by adjusting the motor pulley to deliver the desired air volume

NOTE: Ventilation speed is same as low speed for improved energy savings. Lower operating costs are obtained when the blower is operated on lower speeds.

NOTE: Part-load airflow in cooling mode should not be set below 220 cfm/nominal full load ton to reduce the risk of evaporator coil freeze-up.

MSAV® Multi-Stage Air Volume Sequence of Operation (092 and 102 Models)

- Blower operates in low speed with (G) demand
- Blower operates in low speed for mechanical cooling (Y1) or (Y1+Y2)
- Blower operates in high-speed for any other mode: (Y1+Y2+Y3), and heating
- Economizer damper minimum position is fully closed in unoccupied mode
- In occupied mode, the economizer damper minimum position is determined by the economizer minimum position of the potentiometer
- When outdoor air is not suitable the blower operates on low speed for (Y1) or (Y1+Y2) and switches to high-speed for (Y1+Y2+Y3).
- In free cooling, the blower operates on low speed for (Y1) or (Y1+Y2)

NOTE: Economizer position is determined by mixed air sensor when unit is in free cooling.

MSAV® Multi-Stage Air Volume Operation (120-150 Models)

- MSAV® Multi-Stage Air Volume stages the amount of airflow according to compressor stages, heating demand and ventilation demand
- Units utilize a Variable Frequency Drive (VFD) to stage the supply air blower airflow
- VFD alters the frequency and voltage of the power supply to the blower to control blower speed
- The supply air blower has three speeds:
 1. Low Speed - 1st Stage Cooling
 2. Medium Speed - 2nd Stage Cooling
 3. High Speed - Full load cooling and all heat modes
- Full speed blower operation is set by adjusting the motor pulley to deliver the desired air volume
- Ventilation speed is same as low speed for improved energy savings

NOTE - Part load airflow in cooling mode should not be set below 220 cfm/nominal full load ton to reduce the risk of evaporator coil freeze-up.

- VFD has an operational range of -40 to 125° F outdoor air ambient temperature

NOTE - Lower operating costs are obtained when the blower is operated on lower speeds.

FEATURES AND BENEFITS

BLOWER (continued)

MSAV® Multi-Stage Air Volume Sequence of Operation

- Blower operates in low speed with (G) demand
- Blower operates in low speed for mechanical cooling (Y1)
- Blower operates in medium speed for any other mode (mechanical cooling (Y1+Y2)
- Blower operates in high speed for any other mode (mechanical cooling Y1+Y2+Y3, and heating)
- Economizer damper minimum position is fully closed in unoccupied mode
- In occupied mode, the economizer damper minimum position is determined by the economizer minimum position of the potentiometer

NOTE - Variable Frequency Drive (VFD) is designed to operate on balanced, three-phase power. Operating units on unbalanced three-phase power will reduce the reliability of all electrical components in the unit. Unbalanced power is a result of the power delivery system supplied by the local utility company. Factory-installed inverters are sized to drive blower motors with an equivalent current rating using balanced three-phase power. If unbalanced three-phase power is supplied the installer must replace the existing factory-installed inverter with an inverter that has a higher current rating to allow for the imbalance. Refer to the installation instructions for additional information and replacement information.
Required Selections

Required Selections

- Order blower motor horsepower and drive kit number required when base unit is ordered
- See Drive Kit Specifications Table

CONTROLS

Compressor Monitor

- Two controls are furnished (one for each refrigeration circuit)
- Anti-short cycle control (5 minutes)
- High and low pressure switch monitoring with five-strike lockout protection
- LEDs for diagnostic and troubleshooting
- On-board defrost relays (fan, reversing valve, aux. heat)
- Field selectable frost accumulation timing
- Field selectable compressor delay

Refrigerant Detection System (RDS) Control

- Monitors leak detection sensor
- Terminal strip for thermostat, float switch and RDS alarm connections
- LED for power and sensor status

24V Transformer

- For all control voltage
- Resettable pop-up fuse

Options/Accessories

Field Installed

Smoke Detectors

NOTE - Smoke detectors are not furnished and must be field supplied.

Thermostats

- Control system and thermostat options, see page 10

ELECTRICAL

Marked & Color-Coded Wiring

- All electrical wiring is color-coded and marked to identify which components it is connecting

Electrical Plugs

- Positive connection electrical plugs are used to connect common accessories or maintenance parts for easy removal or installation

Required Selections

Voltage Choice

- Specify when ordering base unit

Options/Accessories

Field Installed

6 Electric Heat

- Helix wound nichrome elements
- Individual element limit controls
- Wiring harness
- See Options / Accessories tables for ordering information

NOTE - Unit Fuse Block is required and must be ordered separately. See Electrical / Electric Heat tables for ordering information.

Bottom Power Entry Kit

- Reduces the number of penetrations in the roof
- Includes bulkhead connectors to provide power and control wiring routing through the roof curb

OPTIONS / ACCESSORIES

INDOOR AIR QUALITY

Air Filters

- Disposable 2 inch MERV 4 filters furnished as standard

Options/Accessories

Field Installed

Replacement Filter Media Kit With Frame

- Replaces existing pleated filter media
- Includes washable metal mesh screen and metal frame with clip for holding replaceable non-pleated filter

Indoor Air Quality (CO₂) Sensors

- Reports to the economizer control to adjust the dampers as needed

ECONOMIZER

Economizer

(Standard and High Performance Common Features)

- Downflow or Horizontal models with Barometric Relief Dampers and Hood
- Barometric Relief Dampers allow relief of excess air
- Aluminum blade dampers prevent blow back and outdoor air infiltration during off cycle
- Exhaust hood with bird screen furnished
- Occupied/Unoccupied mode with field furnished setback thermostat
- Demand Control Ventilation (DCV) ready using optional CO₂ sensors
- Single temperature control is furnished with Economizer
- Outdoor air temperature sensor enables Economizer if the outdoor temperature is less than the setpoint of the control

Field Installed

Standard Economizer Features (Not for Title 24)

- Gear-driven action
- Return air and outdoor air dampers
- Plug-in connections to unit
- Nylon bearings
- Neoprene seals
- 24-volt
- Fully-modulating spring return motor

Standard Economizer Control Module

The Standard Economizer Control Module can be adjusted to operate based on outdoor air temperatures.



Economizer Controls:

- **Damper Minimum Position** - Can be set lower than traditional minimum air requirements resulting in cost savings
- **IAQ Sensor** - Signals dampers to modulate and maintain 55°F when CO₂ is higher than the CO₂ setpoint
- **Demand Control Ventilation (DCV) LED** - A steady green Demand Control Ventilation LED indicates the IAQ reading is higher than setpoint and requires more fresh air
- **Free Cool LED** - A steady green LED indicates outdoor air is suitable for free cooling

NOTE - Free Cooling runs when outdoor air temperature is lower than the set temperature on the economizer control.

NOTE: The Free Cooling default setting for outdoor air temperature sensor is 55°F.

Factory or Field Installed

NOTE - Downflow Economizer is factory or field installed. Horizontal Economizer is field installed only.

High Performance Economizer Features

- Approved for California Title 24 building standards
- Low leakage dampers are Air Movement and Control Association International (AMCA) Class 1A Certified - Maximum 3 CFM per sq. ft. leakage at 1 in. w.g.
- ASHRAE 90.1-2010 compliant
- Gear-driven action
- High torque 24-volt fully-modulating spring return damper motor
- Return air and outdoor air dampers
- Plug-in connections to unit
- Stainless steel bearings
- Enhanced thermoplastic vulcanizate (TPV) seals
- Flexible stainless steel jamb seals minimize air leakage

OPTIONS / ACCESSORIES

ECONOMIZER (continued)

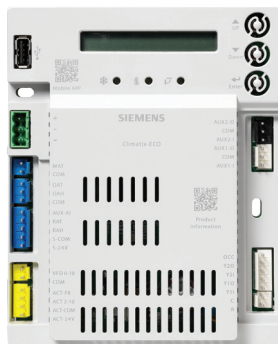
NOTE - Outdoor Air and Barometric Relief Exhaust Hoods are included when economizer is factory installed and are furnished when ordered for field installation.

NOTE - High Performance Economizers are not approved for use with differential enthalpy controls in Title 24 applications.

NOTE - The Free Cooling setpoint for Title 24 applications must be set based on the Climate Zone where the system is installed. See Section 140.4 "Prescriptive Requirements for Space Conditioning Systems" of the California Energy Commission's 2013 Building Energy Efficiency Standards. Refer to Installation Instructions for complete setup information and menu parameters available.

High Performance Economizer Control Module

- Provides inputs and outputs to control economizer based on parameter settings
- Free cooling based on single dry bulb temperature, or combination temperature + humidity sensors
- Automatic switchover for different control modes
- Parameter settings based on climate zone, using GPS functionality in the Climatix Mobile application
- LED indication for free cooling operation, sensor operation and damper operation
- Quick installation and easy commissioning with the **Climatix Mobile App** on a mobile device



NOTE - WLAN Stick is required for App connection to module(s).

- Module displays any alarm messages (fault detection and diagnostics) as an aid in troubleshooting
- RS585 port for BACnet MSTP or Modbus RTU communication
- USB port for firmware updates and WLAN connection for setup and commissioning
- QR codes on module for quick access to download Climatix Mobile App and user documentation
- User Interface for normal operation, parameter setup and, alarm notifications with an LCD display and three operation buttons:

1. **Up Button** - Move to the previous value, step or category
2. **Down Button** - Move to the next value, step or category
3. **Enter Button** -
 - Press to edit the current value or option
 - Press to confirm a newly selected value or option
 - Press Enter + Up to jump up one category
 - Press Enter + Down to jump down one category

NOTE - The Free Cooling setpoint for Title 24 applications must be set based on the Climate Zone where the system is installed. See Section 140.4 "Prescriptive Requirements for Space Conditioning Systems" of the California Energy Commission's 2013 Building Energy Efficiency Standards.

NOTE - Refer to Installation Instructions for complete setup information and menu parameters available.

Field Installed

Single Enthalpy Temperature Control

- Outdoor air enthalpy sensor enables economizer if the outdoor enthalpy is less than the setpoint of the control

Differential Enthalpy Control (Not for Title 24)

- Order one for factory installed economizer
- Order two for field installed economizer
 - One is field installed in the return air section
 - One in the outdoor air section
- Allows the economizer control board to select between outdoor air or return air, whichever has lower enthalpy

WLAN Stick

- Required for Climatix Mobile App usage
- Plugs into USB port on Module to provide a temporary WLAN connection for setup, commissioning, and servicing

NOTE - Only one WLAN Stick is required and can be used on multiple modules.

OPTIONS / ACCESSORIES

EXHAUST

Field Installed

Power Exhaust Fan

- Installs internal to unit for downflow applications only with Economizer option
- Provides exhaust air pressure relief
- Interlocked to run when supply air blower is operating
- Fan runs when outdoor air dampers are 50% open (adjustable)
- Motor is overload protected
- Fan is 20 in. diameter
- Five blades
- One 1/3 HP motor

NOTE - Requires Economizer with Outdoor Air Hood and Barometric Relief Dampers.

OUTDOOR AIR

Field Installed

Outdoor Air Damper

- Downflow or Horizontal
- Linked mechanical dampers
- 0 to 25% (fixed) outdoor air adjustable
- Installs in unit
- Includes outdoor air hood
- Motorized model features fully modulating spring return damper motor with plug-in connection
- Manual model features a slide damper

NOTE - Maximum mixed air temperature in cooling mode: 100°F.

ROOF CURBS

Field Installed

Hybrid Roof Curbs, Downflow

- Nailer strip furnished; mates to unit
- US National Roofing Contractors approved
- Shipped knocked down
- Interlocking tabs fasten corners together
- No tools required for assembly
- Can also be fastened together with furnished hardware
- Available in 8, 14, 18, and 24 inch heights

CEILING DIFFUSERS

Field Installed

(Flush or Step-Down)

- White powder coat finish on diffuser face and grilles
- Insulated UL listed duct liner
- Diffuser box has collars for duct connection
- Step-down diffusers have double deflection blades
- Flush diffusers have fixed blades
- Provisions for suspending
- Internally sealed to prevent recirculation
- Removable return air grille
- Adapts to T-bar ceiling grids or plaster ceilings

Transitions (Supply and Return)

NOTE - Ceiling Diffuser Transitions are not furnished and must be field fabricated.

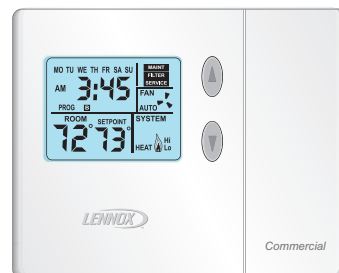
OPTIONAL CONVENTIONAL TEMPERATURE CONTROL SYSTEMS

CS7500 Commercial 7-Day Programmable Thermostat



- Premium Universal Thermostat
- Full Color Touchscreen Interface
- Up To 4 Heat / 3 Cool
- Built-In Sensors For Temperature and Humidity
- Remote Sensors Options For Temperature, Discharge Air, Outdoor Air
- 5-2 or 7-Day Scheduling
- Smooth Setback Recovery
- Heat/Cool Auto-Changeover
- FDD, ASHRAE, IECC Compliant

CS3000 Commercial 5-2 Day Programmable Thermostat



- Conventional Multi-Stage Thermostat
- Intuitive Display
- Push-Button Operation
- Up To 2 Heat / 2 Cool
- Built-In Temperature Sensor
- Remote Temperature Sensing
- Up to 5-2 Day Scheduling
- Smooth Setback Recovery
- Heat/Cool Auto-changeover

Optional Accessories

Cooling Stage-Up Timer Relay

- Allows the unit to attain an additional stage of cooling without the need for extra thermostat connections
- Adjustable - 1 to 1023 seconds
- Mounts internal to unit

OPTIONAL CONVENTIONAL TEMPERATURE CONTROL SYSTEMS

BACnet Compatible Thermostat With Reheat



- 7-Day Programmable
- For units with or without Humiditrol®
- BTL listed MS/TP ensures compatibility with any BACnet system
- Built-in control programs for conventional and heat pump applications
- Conventional systems up to 3-stage heat and 3-stage cool
- Heat pumps with 1 or 2 compressors and up to 2-stage auxiliary heat
- On-board temperature and humidity sensor
- Multiple configurable inputs and outputs enable advanced control strategies
- Set-up Wizard enables rapid system configuration
- No special tools required for installation or commissioning
- Seven-day (2, 4 or 6 event) occupancy scheduling per day
- Backlit 5-inch LCD touchscreen

Description		Order Number
CS7500 Commercial 7-Day Programmable Thermostat		
CS7500 7-Day Thermostat		24K41
Sensors/Accessories	¹ Remote non-adjustable wall-mount 20k	47W36
	¹ Remote non-adjustable wall-mount 10k	47W37
	Remote non-adjustable discharge air (duct mount)	19L22
	Outdoor temperature sensor	X2658
CS3000 5-2 Day Programmable Thermostat		
CS3000 5-2 Day Thermostat		11Y05
Sensor/Accessories	Remote non-adjustable wall mount 10k averaging	47W37
	Thermostat wall mounting plate	X2659
Optional Accessory		
	Cooling Stage-Up Timer Relay	24G40
BACnet 7-Day Programmable Thermostat		
BACnet Controls	² 7-Day BACnet Thermostat	24C57
	³ BACnet Module	16X71
⁴ BACnet Room Sensors	With Display	97W23
	Without Display	97W24
Universal Thermostat Guard with Lock (clear)		
Inside Dimensions (H x W x D) 5-7/8 x 8-3/8 x 3 in.		39P21

¹ Remote wall-mount sensors can be applied in any of the following combinations:
 One Sensor - (1) 47W36, Two Sensors - (2) 47W37, Three Sensors - (2) 47W36 and (1) 47W37
 Four Sensors - (4) 47W36, Five Sensors - (3) 47W36 and (2) 47W37

² BACnet Thermostat (24C57) will control units with and without the Humiditrol® option. If there is a mix of units equipped with and without Humiditrol on the same site, this thermostat can be used for all units if suitable.

³ Not compatible with units equipped with Humiditrol® option.

⁴ Only compatible with BACnet Module (16X71).

OPTIONS / ACCESSORIES

Item Description	Order Number	Size			
		092	102	120	
COOLING/HEATING SYSTEM					
Condensate Drain Trap	PVC	38R23	X	X	X
	Copper	38V21	X	X	X
Corrosion Protection	Factory		O	O	O
Drain Pan Overflow Switch		38A64	X	X	X
Low Ambient Kit		10Z34	X	X	X
BLOWER - SUPPLY AIR					
Blower Motors	Two-Speed Belt Drive - 2 HP	Factory	O	O	
	MSAV® Belt Drive - 3 HP	Factory			O
Drive Kits	Kit #1 590-890 rpm	Factory	O	O	
See Blower Data Tables for selection	Kit #2 800-1105 rpm	Factory	O	O	
	Kit #3 795-1195 rpm	Factory	O	O	
	Kit #4 730-970 rpm	Factory			O
	Kit #5 940-1200 rpm	Factory			O
	Kit #6 1015-1300 rpm	Factory			O
CABINET					
Combination Coil/Hail Guards		12X21	X	X	X

CONTROLS

NOTE - See Conventional Thermostat Control Systems on page 10 for Additional Options.

NOTE - Catalog numbers shown are for ordering field installed accessories.

OX - Configure To Order (Factory Installed) or Field Installed

O = Configure To Order (Factory Installed)

X = Field Installed

OPTIONS / ACCESSORIES

Item Description		Order Number	Size		
			092	102	120
INDOOR AIR QUALITY					
Air Filters					
Replacement Media Filter With Metal Mesh Frame (includes non-pleated filter media)		Y3063	X	X	X
Indoor Air Quality (CO ₂) Sensors					
Sensor - Wall-mount, off-white plastic cover with LCD display		77N39	X	X	X
Sensor - Wall-mount, black plastic case, no display, rated for plenum mounting		23V87	X	X	X
CO ₂ Sensor Duct Mounting Kit - for downflow applications		23Y47	X	X	X
Aspiration Box - For duct mounting non-plenum rated CO ₂ sensors (77N39)		90N43	X	X	X
ELECTRICAL					
Voltage 60 Hz	208/230V - 3 phase	Factory	O	O	O
	460V - 3 phase	Factory	O	O	O
	575V - 3 phase	Factory	O	O	O
Bottom Power Entry Kit		11H66	X	X	X
ELECTRIC HEAT					
7.5 kW	208/240V-3ph	30V24	X	X	
	460V-3ph	30V25	X	X	
	575V-3ph	30V26	X	X	
15 kW	208/240V-3ph	30V30	X	X	X
	460V-3ph	30V31	X	X	X
	575V-3ph	30V32	X	X	X
22.5 kW	208/240V-3ph	30V36	X	X	X
	460V-3ph	30V37	X	X	X
	575V-3ph	30V38	X	X	X
30 kW	208/240V-3ph	30V42	X	X	X
	460V-3ph	30V43	X	X	X
	575V-3ph	30V44	X	X	X
45 kW	208/240V-3ph	30V48	X	X	X
	460V-3ph	30V49	X	X	X
	575V-3ph	30V50	X	X	X
60 kW	208/240V-3ph	30V54			X
	460V-3ph	30V55			X
	575V-3ph	30V56			X
ELECTRIC HEAT ACCESSORIES					
Unit Fuse Block (required) - See Electrical/Electric Heat Tables for Selection			X	X	X

NOTE - Catalog numbers shown are for ordering field installed accessories.

OX - Configure To Order (Factory Installed) or Field Installed

O = Configure To Order (Factory Installed)

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OPTIONS / ACCESSORIES

Item Description	Order Number	Size			
		092	102	120	
ECONOMIZER					
Standard Economizer (Not for Title 24)					
Standard Downflow Economizer with Single Temperature Control - With Barometric Relief Dampers and Air Hoods	24K57	X	X	X	
Standard Horizontal Economizer with Single Temperature Control - With Barometric Relief Dampers and Air Hoods	24K58	X	X	X	
Standard Economizer Controls (Not for Title 24)					
Single Enthalpy Control	21Z09	X	X	X	
Differential Enthalpy Control (order 2)	21Z09	X	X	X	
High Performance Economizer (Approved for California Title 24 Building Standards / AMCA Class 1A Certified)					
High Performance Downflow Economizer with Single Temperature Control - With Barometric Relief Dampers and Air Hoods	24F99	OX	OX	OX	
High Performance Horizontal Economizer with Single Temperature Control - With Barometric Relief Dampers and Air Hoods	24G01	X	X	X	
High Performance Economizer Controls					
Single Enthalpy Control	24G11	X	X	X	
Differential Enthalpy Control (order 2) (Not for Title 24)	24G11	X	X	X	
Economizer Accessories					
WLAN Stick (For High Performance Economizer only)	23K58	X	X	X	
OUTDOOR AIR					
Outdoor Air Dampers					
Motorized Dampers with outdoor air hood	14G36	X	X	X	
Manual Dampers with outdoor air hood	14G37	X	X	X	
POWER EXHAUST					
Standard Static (Downflow)	208/230V-3ph	10Z70	X	X	X
	460V-3ph	10Z71	X	X	X
Standard Static (Horizontal)	208/230V-3ph	24E01	X	X	X
	460V-3ph	28E01	X	X	X
575V Transformer Kit	59E02	X	X	X	
NOTE - Order 575V Transformer Kit with 208/230V Power Exhaust Fan for 575V applications. Order two kits for downflow models, order one kit for horizontal models.					
ROOF CURBS					
Hybrid Roof Curbs, Downflow					
8 in. height	10Z25	X	X	X	
14 in. height	10Z26	X	X	X	
18 in. height	10Z27	X	X	X	
24 in. height	10Z28	X	X	X	
CEILING DIFFUSERS					
Step-Down - Order one	RTD11-95S	13K61	X		
	RTD11-135S	13K62		X	X
Flush - Order one	FD11-95S	13K56	X		
	FD11-135S	13K57		X	X
NOTE - Ceiling Diffuser Transitions are not furnished and must be field fabricated.					
NOTE - Catalog numbers shown are for ordering field installed accessories.					
OX - Configure To Order (Factory Installed) or Field Installed					
O = Configure To Order (Factory Installed)					
X = Field Installed					

SPECIFICATIONS

Model		ZHD092S5T	ZHD102S5T	ZHD120S5M
Nominal Tonnage		7.5	8.5	10
Efficiency Type		Standard	Standard	Standard
Blower Type		Two Speed Belt Drive	Two Speed Belt Drive	MSAV® Multi-Stage Air Volume
Cooling Performance	Gross Cooling Capacity (Btuh)	89,400	103,200	121,900
	¹ Net Cooling Capacity (Btuh)	87,000	100,000	118,000
	¹ AHRI Rated Air Flow (cfm)	2800	3200	3400
	¹ IEER (Btuh/Watt)	14.1	14.1	14.1
	¹ EER (Btuh/Watt)	11.0	11.0	11.0
	Total Unit Power (kW)	7.8	9.1	11.0
Heating Performance	¹ Total High Heating Capacity (Btuh)	83,000	100,000	116,000
	¹ AHRI Rated Air Flow (cfm)	2800	3200	3500
	¹ COP	3.4	3.4	3.4
	Total Unit Power (kW)	7.9	9.0	10.7
	¹ Total Low Heating Capacity (Btuh)	50,000	55,000	70,000
	¹ COP	2.25	2.25	2.25
	Total Unit Power (kW)	7.4	8.6	9.4
	Sound Rating Number	dBA	88	88
Refrigerant Charge	Refrigerant Type	R-454B	R-454B	R-454B
	Circuit 1	14 lbs. 1oz.	14 lbs. 10 oz.	14 lbs. 3 oz.
	Circuit 2	14 lbs. 14 oz.	14 lbs. 13 oz.	12 lbs. 0 oz.
Electric Heat Available - See page 13		7.5, 15, 22.5, 30, 45 kW		7.5, 15, 22.5, 30, 45, 60 kW
Compressor Type (Number)		(1) Two-Stage Scroll, (1) Single-Stage Scroll		
Outdoor Coil	Net face area - ft. ²	26.2	26.2	26.2
	Tube diameter - in.	3/8	3/8	3/8
	Rows	3	3	3
	Fins - in.	20	20	20
Outdoor Coil Fans	Motor HP (number and type)	1/2 (2 PSC)	1/2 (2 PSC)	1/2 (2 PSC)
	Rpm	1075	1075	1075
	Watts	960	960	960
	Diameter (Number) - in.	(2) 24	(2) 24	(2) 24
	Blades	3	3	3
	Total air volume - cfm	9000	9000	9000
	Indoor Coil	Net face area - ft. ²	13.54	13.54
	Tube diameter - in.	3/8	3/8	3/8
	Rows	4	4	4
	Fins - in.	14	14	14
	Condensate drain size (NPT) - in.	(1) 1	(1) 1	(1) 1
	Expansion device type	Balanced Port Thermostatic Expansion Valve		
² Indoor Blower and Drive Selection	Nominal Motor HP	2		3
	Motor - Drive Kit Number	Kit #1		Kit #4
		590-890 rpm		730-970 rpm
		Kit #2		Kit #5
		800-1105 rpm		940-1200 rpm
		Kit #3		Kit #6
	795-1195 rpm	1015-1300 rpm		
Wheel (Number) diameter x width - in.		(1) 15 X 15		
Filters	Type	Disposable		
	Number and size - in.	(4) 20 x 24 x 2		
Line voltage data (Volts-Phase-Hz)		208/230V-3-60 460V-3-60 575V-3-60		

NOTE - Net capacity includes evaporator blower motor heat deduction. Gross capacity does not include evaporator blower motor heat deduction.

¹ AHRI Certified to AHRI Standard 340/360:

Cooling Ratings - 95°F outdoor air temperature and 80°F db/67°F wb entering indoor coil air.

High Temperature Heating Ratings - 47°F db/43°F wb outdoor air temperature and 70°F entering indoor coil air.

Low Temperature Heating Ratings - 17°F db/15°F wb outdoor air temperature and 70°F entering indoor coil air.

² Using total air volume and system static pressure requirements determine from blower performance tables rpm and motor HP required. Maximum usable HP of motors furnished are shown. If motors of comparable HP are used, be sure to keep within the service factor limitations outlined on the motor nameplate.

NOTE - Motor service factor limit - 1.0.

COOLING/HEATING RATINGS

NOTE – For Temperatures and Capacities not shown in tables, see bulletin – Cooling Unit Rating Table Correction Factor Data in Miscellaneous Engineering Data section.

7.5 TON COOLING - ZHD092S5T (1 COMPRESSOR - PART LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																			
		65°F					75°F					85°F					95°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)		
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb		
cfm	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	
63°F	1600	44.1	2.31	0.78	0.98	1	40.8	2.72	0.81	1	1	37.6	3.18	0.84	1	1	34.3	3.69	0.88	0.59	0.59
	1980	46.8	2.28	0.86	1	1	43.7	2.69	0.88	1	1	40.4	3.15	0.93	1	1	36.9	3.66	0.98	0.59	0.59
	2360	49.2	2.26	0.92	1	1	46.1	2.67	0.96	1	1	42.6	3.13	1	1	1	38.9	3.64	1	0.59	0.59
67°F	1600	47.2	2.28	0.57	0.75	0.93	43.7	2.69	0.57	0.78	0.97	40	3.16	0.58	0.81	1	36.1	3.67	0.59	0.59	0.59
	1980	49.2	2.26	0.61	0.83	1	45.6	2.67	0.62	0.85	1	41.8	3.14	0.64	0.89	1	37.7	3.65	0.66	0.59	0.59
	2360	50.6	2.25	0.65	0.9	1	47.2	2.66	0.67	0.93	1	43.2	3.12	0.69	0.98	1	38.7	3.64	0.73	0.59	0.59
71°F	1600	50.5	2.25	0.37	0.54	0.72	47.2	2.66	0.36	0.55	0.74	43.3	3.12	0.34	0.56	0.78	39	3.64	0.35	0.59	0.59
	1980	52.5	2.23	0.38	0.6	0.8	49.1	2.64	0.38	0.61	0.83	44.5	3.11	0.38	0.62	0.86	40.5	3.63	0.36	0.59	0.59
	2360	54	2.22	0.4	0.64	0.87	50.3	2.63	0.4	0.66	0.9	46.1	3.1	0.4	0.69	0.94	41.6	3.62	0.41	0.59	0.59

7.5 TON COOLING - ZHD092S5T (2 COMPRESSORS - PART LOAD / FULL LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																			
		85°F					95°F					105°F					115°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)		
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb		
cfm	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	
63°F	1800	67.6	4.31	0.69	0.82	0.96	62.3	4.98	0.69	0.84	0.98	56.5	5.74	0.7	0.86	1	50.3	6.57	0.71	0.86	0.97
	2250	71.4	4.29	0.74	0.9	1	66	4.97	0.75	0.92	1	60	5.73	0.77	0.95	1	54.1	6.57	0.79	0.93	1
	2700	74.5	4.27	0.79	0.97	1	69	4.96	0.81	0.99	1	63.6	5.71	0.83	1	1	57.8	6.56	0.87	0.99	1
67°F	1800	72.5	4.28	0.53	0.66	0.79	67.1	4.97	0.53	0.67	0.8	61.4	5.72	0.53	0.68	0.83	55.2	6.56	0.52	0.67	0.83
	2250	76.7	4.26	0.57	0.72	0.86	71.1	4.95	0.57	0.73	0.88	64.8	5.71	0.57	0.74	0.91	58.3	6.56	0.57	0.74	0.9
	2700	79.8	4.24	0.6	0.77	0.93	73.8	4.94	0.6	0.79	0.96	67.5	5.7	0.61	0.81	0.99	60.6	6.55	0.62	0.82	0.96
71°F	1800	77.6	4.25	0.4	0.52	0.64	71.9	4.94	0.39	0.52	0.65	66.1	5.71	0.37	0.52	0.65	59.8	6.55	0.36	0.49	0.64
	2250	81.9	4.22	0.41	0.56	0.69	76.1	4.92	0.4	0.56	0.7	70	5.69	0.39	0.56	0.72	63	6.55	0.38	0.54	0.72
	2700	85.1	4.2	0.43	0.59	0.75	79.2	4.91	0.42	0.59	0.76	72.8	5.68	0.41	0.6	0.79	65.6	6.54	0.4	0.59	0.79

7.5 TON COOLING - ZHD092S5T (2 COMPRESSOR - FULL LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																				
		85°F						95°F					105°F					115°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb			
cfm	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F		
63°F	2400	88.3	5.3	0.71	0.86	0.98	82.1	6.01	0.72	0.88	0.99	76.1	6.83	0.73	0.9	1	69.2	7.75	0.75	0.92	1	
	3000	92.7	5.33	0.77	0.94	1	86.7	6.05	0.79	0.96	1	80.3	6.87	0.81	0.97	1	73.6	7.81	0.83	1	1	
	3600	97.1	5.36	0.83	0.98	1	90.7	6.09	0.85	0.99	1	84.4	6.92	0.87	1	1	78.2	7.87	0.91	1	1	
67°F	2400	94.2	5.34	0.55	0.69	0.82	88.3	6.07	0.55	0.7	0.84	81.4	6.89	0.55	0.71	0.86	74.6	7.83	0.55	0.7	0.88	
	3000	99	5.38	0.59	0.75	0.91	92.2	6.11	0.59	0.76	0.93	85.4	6.94	0.6	0.78	0.95	78	7.87	0.61	0.79	0.98	
	3600	102.1	5.42	0.62	0.81	0.96	95.3	6.15	0.63	0.83	0.98	88.1	6.98	0.64	0.85	0.99	80.6	7.91	0.65	0.87	1	
71°F	2400	100.1	5.39	0.41	0.54	0.67	93.7	6.12	0.4	0.54	0.68	87	6.95	0.39	0.54	0.69	79.9	7.89	0.38	0.52	0.68	
	3000	105	5.44	0.43	0.58	0.73	98.4	6.18	0.42	0.58	0.74	91.4	7.01	0.41	0.59	0.76	83.7	7.96	0.41	0.58	0.76	
	3600	108.7	5.47	0.44	0.62	0.79	101.8	6.22	0.44	0.62	0.81	94.3	7.05	0.44	0.64	0.83	86.2	7.99	0.43	0.63	0.84	

7.5 TON HEATING - ZHD092S5T

Indoor Coil Air Volume 70°F Dry Bulb cfm	Air Temperature Entering Outdoor Coil									
	65°F		45°F		25°F		5°F		-15°F	
	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input
	kBtuh	kW	kBtuh	kW	kBtuh	kW	kBtuh	kW	kBtuh	kW
2400	113.7	6.99	89.7	6.30	67.6	5.94	47.6	5.84	29.4	6.03
3000	116.7	6.39	91.2	5.96	68.7	5.81	48.1	5.81	29.7	5.99
2600	118.7	6.04	92.2	5.80	69.3	5.75	48.6	5.79	30.3	5.94

COOLING/HEATING RATINGS

NOTE – For Temperatures and Capacities not shown in tables, see bulletin – Cooling Unit Rating Table Correction Factor Data in Miscellaneous Engineering Data section.

8.5 TON COOLING - ZHD102S5T (1 COMPRESSOR - PART LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																			
		65°F					75°F					85°F					95°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)		
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb		
cfm	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	
63°F	1920	32.6	1.51	0.85	1	1	30.2	1.78	0.87	1	1	27.6	2.08	0.91	1	1	24.9	2.42	0.96	1	1
	2400	34.8	1.49	0.93	1	1	32.3	1.76	0.97	1	1	29.6	2.07	1	1	1	26.7	2.4	1	1	1
	2880	36.3	1.48	1	1	1	33.9	1.75	1	1	1	31.1	2.05	1	1	1	27.9	2.39	1	1	1
67°F	1920	34.4	1.5	0.6	0.81	1	31.7	1.77	0.61	0.84	1	28.7	2.07	0.62	0.88	1	25.6	2.42	0.64	0.93	1
	2400	35.7	1.48	0.65	0.9	1	33	1.76	0.67	0.94	1	30	2.06	0.69	0.99	1	26.6	2.41	0.72	1	1
	2880	36.7	1.47	0.71	0.98	1	34	1.75	0.73	1	1	31.1	2.05	0.76	1	1	27.8	2.39	0.81	1	1
71°F	1920	36.7	1.48	0.37	0.58	0.79	34.1	1.75	0.36	0.59	0.82	31.1	2.05	0.36	0.61	0.85	27.5	2.4	0.35	0.63	0.89
	2400	38.1	1.46	0.39	0.64	0.87	35.2	1.74	0.39	0.66	0.91	32	2.05	0.39	0.68	0.95	28.6	2.39	0.39	0.72	1
	2880	39	1.45	0.42	0.7	0.95	36.1	1.73	0.42	0.72	0.99	32.8	2.04	0.42	0.76	1	29.3	2.38	0.43	0.8	1

8.5 TON COOLING - ZHD102S5T (2 COMPRESSORS - PART LOAD / FULL LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																			
		85°F					95°F					105°F					115°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)		
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb		
				75°F	80°F	85°F			75°F	80°F	85°F			75°F	80°F	85°F			75°F	80°F	85°F
cfm	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	
63°F	2040	67.4	3.39	0.71	0.86	1	62.1	3.92	0.72	0.88	1	56.5	4.52	0.74	0.91	1	50.3	5.18	0.76	0.95	1
	2550	71.1	3.37	0.77	0.95	1	65.7	3.91	0.79	0.97	1	60.1	4.51	0.81	1	1	54.6	5.17	0.84	1	1
	3060	74.7	3.35	0.83	1	1	69.5	3.9	0.86	1	1	64.1	4.5	0.89	1	1	58.3	5.17	0.92	1	1
67°F	2040	72.4	3.36	0.55	0.69	0.83	67	3.91	0.55	0.7	0.85	61.1	4.51	0.55	0.71	0.87	54.9	5.17	0.55	0.73	0.91
	2550	76.3	3.35	0.59	0.75	0.91	70.5	3.89	0.59	0.77	0.94	64.4	4.5	0.59	0.79	0.97	57.8	5.17	0.6	0.82	1
	3060	78.9	3.33	0.62	0.81	0.98	72.9	3.88	0.63	0.83	1	66.7	4.49	0.64	0.86	1	59.9	5.16	0.65	0.9	1
71°F	2040	77.5	3.34	0.41	0.54	0.67	71.9	3.89	0.39	0.54	0.68	66	4.49	0.38	0.54	0.69	59.6	5.16	0.37	0.54	0.71
	2550	81.5	3.32	0.42	0.58	0.73	75.7	3.87	0.41	0.58	0.74	69.5	4.48	0.4	0.59	0.76	62.6	5.16	0.39	0.59	0.79
	3060	84.3	3.3	0.44	0.62	0.79	78.4	3.86	0.43	0.62	0.81	71.6	4.48	0.42	0.63	0.84	64.7	5.15	0.42	0.65	0.87

8.5 TON COOLING - ZHD102S5T (2 COMPRESSORS - FULL LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																				
		85°F						95°F					105°F					115°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb			
cfm	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F		
63°F	2720	92.6	5.33	0.74	0.9	1	86.6	6.05	0.76	0.92	1	79.7	6.87	0.77	0.95	1	73.3	7.8	0.79	0.97	1	
	3400	98.1	5.38	0.81	0.97	1	91.3	6.1	0.83	0.98	1	84.8	6.92	0.85	1	1	78.6	7.88	0.88	1	1	
	4080	102	5.41	0.87	1	1	96.2	6.15	0.9	1	1	89.4	6.98	0.93	1	1	82.6	7.93	0.95	1	1	
67°F	2720	99	5.38	0.57	0.72	0.87	92.4	6.11	0.57	0.73	0.89	85.7	6.94	0.58	0.75	0.92	78.2	7.87	0.58	0.77	0.95	
	3400	103.2	5.43	0.61	0.79	0.95	96.6	6.16	0.62	0.81	0.96	89.1	6.98	0.63	0.83	0.98	81.5	7.93	0.64	0.86	1	
	4080	106.3	5.46	0.65	0.85	0.99	99.3	6.19	0.66	0.88	1	91.8	7.02	0.68	0.91	1	83.9	7.96	0.69	0.94	1	
71°F	2720	105	5.44	0.42	0.56	0.7	98.4	6.17	0.41	0.56	0.71	91.5	7.01	0.41	0.57	0.73	84.1	7.96	0.4	0.57	0.75	
	3400	109.8	5.48	0.44	0.6	0.77	102.9	6.23	0.43	0.61	0.79	95.5	7.06	0.43	0.62	0.81	87.6	8.01	0.42	0.63	0.84	
	4080	113.2	5.52	0.46	0.65	0.83	105.9	6.26	0.45	0.66	0.86	97.7	7.09	0.45	0.67	0.89	89.8	8.04	0.45	0.69	0.92	

8.5 TON HEATING - ZHD102S5T

Indoor Coil Air Volume 70°F Dry Bulb cfm	Air Temperature Entering Outdoor Coil									
	65°F		45°F		25°F		5°F		-15°F	
	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input
	kBtuh	kW	kBtuh	kW	kBtuh	kW	kBtuh	kW	kBtuh	kW
2720	129.7	7.83	101.9	7.01	77.5	6.55	54.5	6.53	32.8	6.80
3400	133.0	7.17	103.9	6.62	78.7	6.42	55.3	6.49	33.4	6.76
4080	135.3	6.78	105.2	6.43	79.5	6.38	56.1	6.47	34.2	6.72

COOLING/HEATING RATINGS

NOTE – For Temperatures and Capacities not shown in tables, see bulletin – Cooling Unit Rating Table Correction Factor Data in Miscellaneous Engineering Data section.

10 TON COOLING - ZHD120S5M (1 COMPRESSOR - PART LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																			
		65°F					75°F					85°F					95°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)		
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb		
cfm	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	
63°F	2110	44.9	2.33	0.89	1	1	41.9	2.73	0.93	1	1	38.7	3.18	0.97	1	1	35.3	3.69	1	1	1
	2640	47.6	2.31	0.99	1	1	44.6	2.7	1	1	1	41.2	3.15	1	1	1	37.5	3.66	1	1	1
	3170	49.6	2.28	1	1	1	46.4	2.69	1	1	1	42.7	3.14	1	1	1	39	3.65	1	1	1
67°F	2110	46.7	2.31	0.63	0.86	1	43.3	2.72	0.65	0.9	1	39.5	3.17	0.67	0.94	1	35.6	3.68	0.7	1	1
	2640	48.3	2.3	0.69	0.96	1	44.7	2.7	0.71	1	1	41.3	3.15	0.74	1	1	37.3	3.67	0.79	1	1
	3170	49.7	2.28	0.75	1	1	46.3	2.69	0.78	1	1	42.7	3.14	0.82	1	1	39	3.65	0.87	1	1
71°F	2110	49.8	2.28	0.39	0.62	0.83	46.2	2.69	0.38	0.64	0.87	42.3	3.15	0.39	0.66	0.91	38.2	3.66	0.39	0.69	0.82
	2640	51.4	2.26	0.41	0.68	0.93	47.8	2.67	0.42	0.71	0.97	43.9	3.13	0.42	0.74	1	39.6	3.64	0.43	0.78	0.92
	3170	52.7	2.25	0.45	0.75	1	48.9	2.66	0.44	0.78	1	44.8	3.12	0.45	0.82	1	40.5	3.63	0.48	0.87	0.99

10 TON COOLING - ZHD120S5M (2 COMPRESSORS - PART LOAD / FULL LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																				
		85°F						95°F					105°F					115°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb			
cfm	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F	kBtuh	kW	75°F	80°F	85°F		
63°F	2400	108.8	6.73	0.75	0.89	0.98	101.8	7.75	0.76	0.9	0.99	93.8	8.89	0.78	0.92	1	86.2	10.2	0.81	0.94	0.96	
	3000	114.8	6.76	0.81	0.94	1	107.6	7.78	0.83	0.96	1	98.9	8.92	0.85	0.98	1	90.8	10.23	0.87	1	1	
	3600	119.4	6.77	0.87	0.99	1	111.8	7.8	0.88	1	1	104	8.96	0.91	1	1	95.6	10.28	0.92	1	1	
67°F	2400	115.1	6.77	0.58	0.72	0.86	107.1	7.79	0.58	0.74	0.88	98.8	8.94	0.59	0.76	0.89	90.1	10.25	0.59	0.78	0.83	
	3000	120.1	6.81	0.62	0.79	0.92	111.9	7.83	0.62	0.81	0.94	103.5	8.99	0.64	0.83	0.96	94.7	10.29	0.65	0.85	0.89	
	3600	124.2	6.84	0.65	0.85	0.97	116	7.86	0.67	0.87	0.99	107.2	9.02	0.68	0.89	1	97.9	10.32	0.71	0.91	0.95	
71°F	2400	122.4	6.81	0.42	0.56	0.7	114.4	7.84	0.42	0.57	0.72	105.7	9	0.41	0.58	0.74	96.6	10.32	0.41	0.59	0.66	
	3000	127.8	6.86	0.44	0.61	0.77	119.2	7.89	0.44	0.62	0.79	109.8	9.05	0.43	0.63	0.81	100.4	10.36	0.43	0.65	0.73	
	3600	131.9	6.89	0.46	0.65	0.83	122.7	7.92	0.46	0.66	0.85	113.3	9.08	0.46	0.68	0.87	103	10.38	0.46	0.7	0.8	

10 TON COOLING - ZHD120S5M (2 COMPRESSORS - FULL LOAD)

Entering Wet Bulb Temperature	Total Air Volume	Outdoor Air Temperature Entering Outdoor Coil																			
		85°F					95°F					105°F					115°F				
		Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)			Total Cool Cap.	Comp. Motor Input	Sensible To Total Ratio (S/T)		
				Dry Bulb					Dry Bulb					Dry Bulb					Dry Bulb		
cfm	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	kBtu/h	kW	75°F	80°F	85°F	
63°F	3200	122.4	7.95	0.71	0.85	0.96	112.3	9.03	0.72	0.87	0.98	105.8	10.26	0.74	0.9	1	97.5	11.68	0.75	0.92	1
	4000	128.4	8.02	0.77	0.92	1	114.3	9.11	0.78	0.95	1	112.4	10.36	0.8	0.96	1	102.8	11.77	0.83	0.99	1
	4800	133.9	8.09	0.82	0.97	1	116.2	9.18	0.84	0.99	1	116.9	10.43	0.87	1	1	108.3	11.86	0.9	1	1
67°F	3200	130	8.04	0.56	0.69	0.82	118.7	9.13	0.56	0.7	0.84	112.9	10.37	0.56	0.71	0.86	103.7	11.78	0.57	0.73	0.89
	4000	136	8.12	0.59	0.74	0.89	122.3	9.21	0.59	0.76	0.92	118.2	10.45	0.6	0.78	0.94	108.4	11.86	0.61	0.81	0.97
	4800	140.7	8.19	0.62	0.8	0.95	124.5	9.28	0.63	0.82	0.97	121.8	10.51	0.64	0.85	0.99	111.8	11.92	0.66	0.88	1
71°F	3200	137.9	8.15	0.42	0.54	0.66	125.5	9.24	0.41	0.54	0.68	120.8	10.49	0.4	0.55	0.69	111.3	11.91	0.4	0.56	0.7
	4000	144.9	8.24	0.43	0.58	0.72	126.6	9.33	0.42	0.58	0.74	126.3	10.59	0.42	0.59	0.76	115.6	11.99	0.42	0.61	0.78
	4800	149.7	8.31	0.44	0.61	0.78	129.3	9.39	0.44	0.62	0.8	129.4	10.64	0.44	0.64	0.83	119.1	12.05	0.44	0.65	0.85

10 TON HEATING - ZHD120S5M

Indoor Coil Air Volume 70°F Dry Bulb cfm	Air Temperature Entering Outdoor Coil									
	65°F		45°F		25°F		5°F		-15°F	
	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input	Total Heating Capacity	Comp. Motor Input
	kBtuh	kW	kBtuh	kW	kBtuh	kW	kBtuh	kW	kBtuh	kW
3200	153.1	9.38	120.1	8.32	90.8	7.69	65.2	7.39	40.1	7.36
4000	156.2	8.62	121.6	7.87	91.9	7.48	66.0	7.32	40.9	7.33
4800	158.7	8.17	123.1	7.64	92.8	7.37	66.8	7.30	41.8	7.30

BLOWER DATA

BASE UNIT

BLOWER TABLE INCLUDES RESISTANCE FOR BASE UNIT ONLY (NO HEAT SECTION) WITH DRY INDOOR COIL AND AIR FILTERS IN PLACE. FOR ALL UNITS ADD:

1 – Wet indoor coil air resistance of selected unit.

2 – Any factory installed options air resistance (heat section, economizer, etc.)

3 – Any field installed accessories air resistance (duct resistance, diffuser, etc.)

Then determine from blower table blower motor output required.

See page 20 for blower motors and drives and air resistance for wet coil and options/accessories.

Minimum Air Volume Required For Use With Optional Electric Heat (Maximum Static Pressure - 2.0 in. w.g.)

15 kW, 22.5 kW - 2065 cfm; 30 kW - 2250 cfm; 45 kW - 3000 cfm; 60 kW - 4000 cfm

Total Air Volume cfm	Total Static Pressure – in. w.g.																											
	0.2		0.4		0.6		0.8		1.0		1.2		1.4		1.6		1.8		2		2.2		2.4		2.6			
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP		
2000	542	0.43	602	0.60	664	0.75	732	0.89	802	1.02	869	1.15	927	1.27	979	1.41	1029	1.57	1079	1.75	1129	1.95	1179	2.15	1230	2.37		
2250	560	0.55	619	0.71	681	0.86	748	1.00	817	1.14	882	1.27	939	1.41	991	1.57	1041	1.74	1090	1.93	1140	2.13	1190	2.35	1241	2.57		
2500	579	0.68	637	0.83	699	0.98	766	1.12	834	1.26	897	1.41	953	1.57	1005	1.74	1054	1.92	1103	2.12	1152	2.33	1202	2.55	1254	2.79		
2750	599	0.81	657	0.97	719	1.11	785	1.25	851	1.41	913	1.57	968	1.74	1020	1.93	1068	2.13	1116	2.34	1165	2.56	1215	2.78	1268	3.01		
3000	620	0.95	678	1.11	741	1.25	806	1.40	870	1.58	930	1.75	985	1.94	1036	2.14	1084	2.36	1131	2.58	1180	2.80	1230	3.02	1283	3.26		
3250	643	1.10	701	1.26	764	1.41	828	1.57	891	1.76	950	1.95	1003	2.16	1053	2.38	1100	2.61	1148	2.83	1196	3.06	1246	3.29	1299	3.52		
3500	667	1.26	726	1.43	788	1.58	851	1.77	913	1.97	970	2.17	1023	2.41	1071	2.65	1118	2.88	1165	3.11	1213	3.33	1264	3.57	1317	3.81		
3750	693	1.44	752	1.61	813	1.78	876	1.98	936	2.20	992	2.43	1043	2.68	1091	2.93	1137	3.17	1183	3.40	1232	3.64	1284	3.88	1338	4.13		
4000	720	1.65	779	1.82	840	2.00	902	2.22	961	2.46	1015	2.71	1064	2.98	1111	3.24	1156	3.48	1203	3.72	1253	3.96	1305	4.22	1359	4.48		
4250	748	1.86	807	2.04	868	2.24	929	2.48	986	2.75	1038	3.02	1086	3.30	1132	3.57	1177	3.81	1224	4.05	1274	4.31	1327	4.57	1382	4.85		
4500	778	2.09	837	2.28	898	2.51	957	2.78	1012	3.07	1062	3.37	1108	3.65	1154	3.92	1199	4.17	1247	4.41	1297	4.67	1350	4.94	1405	5.22		
4750	809	2.34	868	2.56	929	2.82	986	3.12	1038	3.43	1087	3.74	1132	4.03	1177	4.29	1223	4.54	1270	4.79	1321	5.04	1374	5.31	1428	5.58		
5000	841	2.62	901	2.87	960	3.17	1015	3.50	1065	3.83	1112	4.14	1157	4.43	1201	4.69	1247	4.94	1295	5.18	1345	5.42	1398	5.68	---	---		
5250	875	2.93	935	3.23	992	3.56	1044	3.91	1092	4.26	1138	4.57	1182	4.85	1226	5.10	1272	5.34	1320	5.57	---	---	---	---	---	---		
5500	911	3.30	969	3.63	1024	4.00	1074	4.37	1120	4.71	1165	5.02	1208	5.29	1253	5.53	---	---	---	---	---	---	---	---	---	---		
5750	948	3.71	1004	4.08	1056	4.48	1104	4.85	1148	5.19	1192	5.49	1235	5.74	---	---	---	---	---	---	---	---	---	---	---	---		
6000	985	4.18	1039	4.59	1088	5.00	1134	5.37	1177	5.69	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---		
6250	1022	4.70	1073	5.14	1120	5.54	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---		

BLOWER DATA

FACTORY INSTALLED BELT DRIVE KIT SPECIFICATIONS

Nominal HP	Drive Kit Number	RPM Range
2	1	590 - 890
2	2	800 - 1105
2	3	795 - 1195
3	4	730 - 970
3	5	940 - 1200
3	6	1015 - 1300

NOTE - Using total air volume and system static pressure requirements determine from blower performance tables rpm and motor output required. Maximum usable output of motors furnished are shown. If motors of comparable output are used, be sure to keep within the service factor limitations outlined on the motor nameplate.

NOTE – Motor service factor limit - 1.0.

POWER EXHAUST FAN PERFORMANCE

Return Air System Static Pressure	Air Volume Exhausted
in. w.g.	cfm
0	3575
0.05	3405
0.10	3550
0.15	3245
0.20	3115
0.25	3020
0.30	2900
0.35	2785

FACTORY INSTALLED OPTIONS/FIELD INSTALLED ACCESSORY AIR RESISTANCE - in. w.g.

Air Volume cfm	Wet Indoor Coil	Electric Heat	Economizer	Filters	
				MERV 8	MERV 13
1750	0.04	0.03	0.03	0.01	0.03
2000	0.05	0.03	0.05	0.01	0.03
2250	0.06	0.04	0.06	0.01	0.04
2500	0.07	0.04	0.08	0.01	0.05
2750	0.08	0.05	0.09	0.02	0.05
3000	0.09	0.06	0.11	0.02	0.06
3250	0.10	0.06	0.13	0.02	0.06
3500	0.11	0.09	0.15	0.03	0.07
3750	0.13	0.09	0.17	0.03	0.08
4000	0.14	0.09	0.19	0.04	0.08
4250	0.15	0.13	0.21	0.04	0.09
4500	0.17	0.14	0.24	0.04	0.09
4750	0.18	0.17	0.26	0.05	0.10
5000	0.20	0.20	0.29	0.06	0.10
5250	0.22	0.22	0.32	0.06	0.11
5500	0.23	0.25	0.34	0.07	0.12
5750	0.25	0.31	0.37	0.07	0.12
6000	0.27	0.33	0.40	0.08	0.13

BLOWER DATA

CEILING DIFFUSERS AIR RESISTANCE - in. w.g.

Size	RTD11 Step-Down Diffuser				FD11 Flush Diffuser
	Air Volume cfm	2 Ends Open	1 Side, 2 Ends Open	All Ends & Sides Open	
092	2400	0.21	0.18	0.15	0.14
	2600	0.24	0.21	0.18	0.17
	2800	0.27	0.24	0.21	0.20
	3000	0.32	0.29	0.25	0.25
	3200	0.41	0.37	0.32	0.31
	3400	0.50	0.45	0.39	0.37
	3600	0.61	0.54	0.48	0.44
	3800	0.73	0.63	0.57	0.51
102, 120	3600	0.36	0.28	0.23	0.15
	3800	0.40	0.32	0.26	0.18
	4000	0.44	0.36	0.29	0.21
	4200	0.49	0.40	0.33	0.24
	4400	0.54	0.44	0.37	0.27
	4600	0.60	0.49	0.42	0.31
	4800	0.65	0.53	0.46	0.35
	5000	0.69	0.58	0.50	0.39
	5200	0.75	0.62	0.54	0.43

CEILING DIFFUSER AIR THROW DATA

Size	Air Volume	¹ Effective Throw Range	
		RTD11 Step-Down	FD11 Flush
	cfm	ft.	ft.
092	2600	24 - 29	19 - 24
	2800	25 - 30	20 - 28
	3000	27 - 33	21 - 29
	3200	28 - 35	22 - 29
	3400	30 - 37	22 - 30
102, 120	3600	25 - 33	22 - 29
	3800	27 - 35	22 - 30
	4000	29 - 37	24 - 33
	4200	32 - 40	26 - 35
	4400	34 - 42	28 - 37

¹ Throw is the horizontal or vertical distance an air stream travels on leaving the outlet or diffuser before the maximum velocity is reduced to 50 ft. per minute. Four sides open.

ELECTRICAL/ELECTRIC HEAT DATA

7.5 TON

Model		ZHD092S5T		
¹ Voltage - 60Hz		208/230V - 3 Ph	460V - 3 Ph	575V - 3 Ph
Compressor 1	Rated Load Amps	11.9	6.8	4.8
	Locked Rotor Amps	112	61.8	39
Compressor 2	Rated Load Amps	12.2	6.4	5.1
	Locked Rotor Amps	120	50	41
Outdoor Fan Motors (2)	Full Load Amps (2 Non-ECM)	3	1.5	1.2
	Total	6	3	2.4
Power Exhaust (2) 0.33 HP	Full Load Amps	2.4	1.3	1
	Total	4.8	2.6	2
Service Outlet 115V GFI (amps)		15	15	20
Indoor Blower Motor	Horsepower	2	2	2
	Full Load Amps	7.5	3.4	2.7
² Maximum Overcurrent Protection	Unit Only	50	25	20
	With (2) 0.50 HP Power Exhaust	50	30	20
³ Minimum Circuit Ampacity	Unit Only	41	22	17
	With (2) 0.33 HP Power Exhaust	46	24	19

ELECTRIC HEAT DATA

Electric Heat Voltage			208V	240V	480V	600V
² Maximum Overcurrent Protection	Unit+	7.5 kW	70	70	35	30
	Electric Heat	15 kW	80	90	45	35
		22.5 kW	100	110	60	45
		30 kW	125	150	70	60
		45 kW	175	200	90	80
³ Minimum Circuit Ampacity	Unit+	7.5 kW	61	64	33	26
	Electric Heat	15 kW	80	86	44	35
		22.5 kW	100	109	56	44
		30 kW	119	131	67	53
		45 kW	158	176	89	71
² Maximum Overcurrent Protection	Unit+	7.5 kW	70	70	40	30
	Electric Heat and (2) 0.33 HP Power Exhaust	15 kW	90	100	50	40
		22.5 kW	110	125	60	50
		30 kW	125	150	70	60
		45 kW	175	200	100	80
³ Minimum Circuit Ampacity	Unit+	7.5 kW	65	69	36	28
	Electric Heat and (2) 0.33 HP Power Exhaust	15 kW	85	91	47	37
		22.5 kW	105	114	58	46
		30 kW	124	136	70	55
		45 kW	163	181	92	73

ELECTRIC HEAT ACCESSORIES

Unit Fuse Block	Unit Only	11M12	11M12	11M10	11M09
	Unit + Power Exhaust	11M12	11M12	11M10	11M09

NOTE - All units have a minimum Short Circuit Current Rating (SCCR) of 5000 amps.

¹ Extremes of operating range are plus and minus 10% of line voltage.² HACR type breaker or fuse.³ Refer to National or Canadian Electrical Code manual to determine wire, fuse and disconnect size requirements.

ELECTRICAL/ELECTRIC HEAT DATA

8.5 TON

Model		ZHD102S5T		
¹ Voltage - 60Hz		208/230V - 3 Ph	460V - 3 Ph	575V - 3 Ph
Compressor 1	Rated Load Amps	13.8	6.9	5.8
	Locked Rotor Amps	150	58	47.8
Compressor 2	Rated Load Amps	12.2	6.4	5.1
	Locked Rotor Amps	120	50	41
Outdoor Fan Motors (2)	Full Load Amps (2 Non-ECM)	3	1.5	1.2
	Total	6	3	2.4
Power Exhaust (2) 0.33 HP	Full Load Amps	2.4	1.3	1
	Total	4.8	2.6	2
Service Outlet 115V GFI (amps)		15	15	20
Indoor Blower Motor	Horsepower	2	2	2
	Full Load Amps	7.5	3.4	2.7
² Maximum Overcurrent Protection	Unit Only	50	25	20
	With (2) 0.33 HP Power Exhaust	60	30	25
³ Minimum Circuit Ampacity	Unit Only	43	22	18
	With (2) 0.33 HP Power Exhaust	48	25	20

ELECTRIC HEAT DATA

Electric Heat Voltage			208V	240V	480V	600V
² Maximum Overcurrent Protection	Unit+	7.5 kW	70	70	35	30
	Electric Heat	15 kW	90	90	45	40
		22.5 kW	110	125	60	45
		30 kW	125	150	70	60
		45 kW	175	200	90	80
³ Minimum Circuit Ampacity	Unit+	7.5 kW	63	66	33	27
	Electric Heat	15 kW	83	89	44	36
		22.5 kW	102	111	56	45
		30 kW	122	134	67	54
		45 kW	161	179	90	72
² Maximum Overcurrent Protection	Unit+	7.5 kW	70	80	40	30
	Electric Heat and (2) 0.33 HP Power Exhaust	15 kW	90	100	50	40
		22.5 kW	110	125	60	50
		30 kW	150	150	70	60
		45 kW	175	200	100	80
³ Minimum Circuit Ampacity	Unit+	7.5 kW	68	71	36	29
	Electric Heat and (2) 0.33 HP Power Exhaust	15 kW	87	93	47	38
		22.5 kW	107	116	58	47
		30 kW	126	138	70	56
		45 kW	166	184	92	74

ELECTRIC HEAT ACCESSORIES

Unit Fuse Block	Unit Only	11M12	11M12	11M10	11M09
	Unit + Power Exhaust	11M12	11M12	11M10	11M09

NOTE - All units have a minimum Short Circuit Current Rating (SCCR) of 5000 amps.

¹ Extremes of operating range are plus and minus 10% of line voltage.² HACR type breaker or fuse.³ Refer to National or Canadian Electrical Code manual to determine wire, fuse and disconnect size requirements.

ELECTRICAL/ELECTRIC HEAT DATA

10 TON

Model		ZHD120S5M		
¹ Voltage - 60Hz		208/230V - 3 Ph	460V - 3 Ph	575V - 3 Ph
Compressor 1	Rated Load Amps	13.8	6.9	5.8
	Locked Rotor Amps	150	58	47.8
Compressor 2	Rated Load Amps	19.6	8.2	6.6
	Locked Rotor Amps	136	66.1	55.3
Outdoor Fan Motors (2)	Full Load Amps (2 Non-ECM)	3	1.5	1.2
	Total	6	3	2.4
Power Exhaust (2) 0.33 HP	Full Load Amps	4.4	1.7	1.7
	Total	8.8	3.4	3.4
Service Outlet 115V GFI (amps)		15	15	20
Indoor Blower Motor	Horsepower	3	3	3
	Full Load Amps	10.6	4.8	3.9
² Maximum Overcurrent Protection	Unit Only	45	20	15
	With (2) 0.33 HP Power Exhaust	50	25	20
³ Minimum Circuit Ampacity	Unit Only	34	17	14
	With (2) 0.33 HP Power Exhaust	39	20	16

ELECTRIC HEAT DATA

Electric Heat Voltage			208V	240V	480V	600V
² Maximum Overcurrent Protection	Unit+ Electric Heat	15 kW	80	80	40	35
		22.5 kW	100	110	60	45
		30 kW	125	125	70	50
		45 kW	175	175	90	70
		60 kW	175	200	90	80
³ Minimum Circuit Ampacity	Unit+ Electric Heat	15 kW	73	79	39	32
		22.5 kW	93	102	51	41
		30 kW	113	125	62	50
		45 kW	152	170	85	68
		60 kW	159	179	89	72
² Maximum Overcurrent Protection	Unit+ Electric Heat and (2) 0.33 HP Power Exhaust	15 kW	80	90	45	35
		22.5 kW	100	110	60	45
		30 kW	125	150	70	60
		45 kW	175	175	90	70
		60 kW	175	200	100	80
³ Minimum Circuit Ampacity	Unit+ Electric Heat and (2) 0.33 HP Power Exhaust	15 kW	78	84	42	34
		22.5 kW	98	107	53	43
		30 kW	117	129	65	52
		45 kW	156	174	87	70
		60 kW	164	183	92	74

ELECTRIC HEAT ACCESSORIES

Unit Fuse Block	Unit Only	11M12	11M12	11M10	11M09
	Unit + Power Exhaust	11M12	11M12	11M10	11M09

NOTE - All units have a minimum Short Circuit Current Rating (SCCR) of 5000 amps.

¹ Extremes of operating range are plus and minus 10% of line voltage.² HACR type breaker or fuse.³ Refer to National or Canadian Electrical Code manual to determine wire, fuse and disconnect size requirements.

ELECTRIC HEAT CAPACITIES

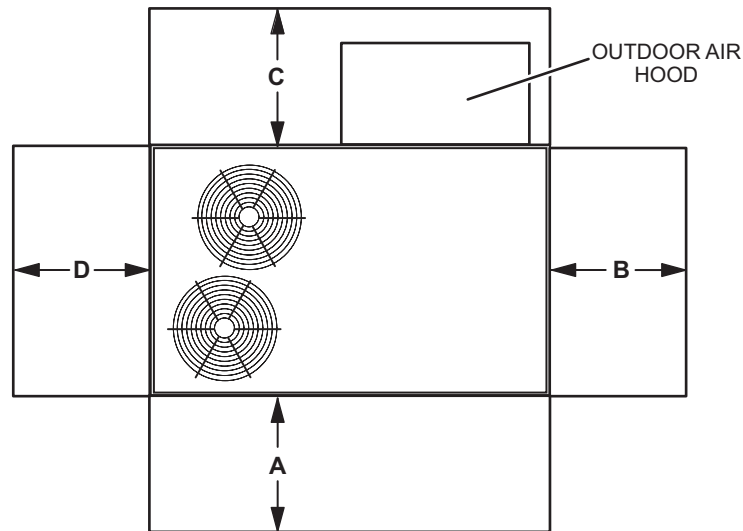
Volts Input	7.5 kW			15 kW			22.5 kW			30 kW			45 kW			60 kW		
	kW Input	Btuh Output	No. of Stages	kW Input	Btuh Output	No. of Stages	kW Input	Btuh Output	No. of Stages	kW Input	Btuh Output	No. of Stages	kW Input	Btuh Output	No. of Stages	kW Input	Btuh Output	No. of Stages
208	5.6	19,100	1	11.3	38,600	1	16.9	57,700	1	22.5	76,800	1	33.8	115,300	1	45.0	153,600	1
220	6.3	21,500	1	12.6	43,000	1	18.9	64,500	1	25.2	86,000	1	37.8	129,000	1	50.4	172,000	1
230	6.9	23,600	1	13.8	47,100	1	20.7	70,700	1	27.5	93,900	1	41.3	141,000	1	55.1	188,000	1
240	7.5	25,600	1	15.0	51,200	1	22.5	76,800	1	30.0	102,400	1	45.0	153,600	1	60.0	204,800	1
440	6.9	21,500	1	12.6	43,000	1	18.9	64,500	1	25.2	86,000	1	37.8	129,000	1	50.4	172,000	1
460	6.9	23,600	1	13.8	47,100	1	20.7	70,700	1	27.5	93,900	1	41.3	141,000	1	55.1	188,000	1
480	7.5	25,600	1	15.0	51,200	1	22.5	76,800	1	30.0	102,400	1	45.0	153,600	1	60.0	204,800	1
550	6.3	21,500	1	12.6	43,000	1	18.9	64,500	1	25.2	86,000	1	37.8	129,000	1	50.4	172,000	1
575	6.9	23,600	1	13.8	47,100	1	20.7	70,700	1	27.5	93,900	1	41.3	141,000	1	55.1	188,000	1
600	7.5	25,600	1	15.0	51,200	1	22.5	76,800	2	30.0	102,400	2	45.0	153,600	2	60.0	204,800	1

FIELD WIRING NOTES

- For use with copper wiring only
- Field wiring not furnished
- All wiring must conform to NEC or CEC and local electrical codes
- For specific wiring information, please refer to the installation instructions

UNIT CLEARANCES

UNIT WITH ECONOMIZER



¹ Unit Clearance	A		B		C		D		Top Clearance
	in.	mm	in.	mm	in.	mm	in.	mm	
Service Clearance	60	1524	36	914	36	914	60	1524	Unobstructed
Minimum Operation Clearance	36	914	36	914	36	914	36	914	

NOTE - Entire perimeter of unit base requires support when elevated above the mounting surface.

¹ Service Clearance - Required for removal of serviceable parts.

Minimum Operation Clearance - Required clearance for proper unit operation.

OUTDOOR SOUND DATA

Size	Octave Band Sound Power Levels dBA, re 10 ⁻¹² Watts - Center Frequency - Hz							¹ Sound Rating Number (dBA)
	125	250	500	1000	2000	4000	8000	
092, 102	76	79	84	83	79	73	66	88
120	77	80	85	84	79	74	66	88

¹ Sound Rating Number according to AHRI Standard 270-2008. Sound Rating Number is the overall A-Weighted Sound Power Level, (LWA), dB (100 Hz to 10,000 Hz).

WEIGHT DATA

Size	Net		Shipping	
	lbs.	kg	lbs.	kg
092S Base Unit	1068	484	1153	523
092S Max. Unit	1225	556	1310	594
102S Base Unit	1068	484	1153	523
102S Max. Unit	1225	556	1310	594
120S Base Unit	1126	511	1211	549
120S Max. Unit	1290	585	1375	624

FACTORY / FIELD INSTALLED OPTIONS AND ACCESSORIES - NET WEIGHTS

Description	lbs.	kg
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ECONOMIZER / OUTDOOR AIR / POWER EXHAUST

Economizer - With Barometric Relief Dampers and Hoods

Downflow	90	41
Horizontal	95	43
Horizontal Low Profile Barometric Relief Dampers with Hood	8	4

Outdoor Air Dampers With Hood

Morotized	44	20
Manual	27	12

Power Exhaust

Downflow	60	27
Horizontal	41	19

ELECTRIC HEAT

7.5 kW	90	41
15 kW	90	41
22.5 kW	90	41
30 kW	90	41
45 kW	90	41
60 kW	90	41

COIL/HAIL GUARDS

All models	50	23
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ROOF CURBS

Hybrid Roof Curbs, Downflow

8 in. height	79	36
14 in. height	104	47
18 in. height	120	54
24 in. height	145	66

CEILING DIFFUSERS

Step-Down	RTD11-95S	118	54
	RTD11-135S	135	61
Flush	FD11-95S	118	54
	FD11-135S	135	61

DIMENSIONS		UNIT
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UNIT

Size	CORNER WEIGHTS																CENTER OF GRAVITY							
	AA				BB				CC				DD				EE				FF			
	Base		Max.		Base		Max.		Base		Max.		Base		Max.		Base		Max.		Base		Max.	
	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	in.	mm	in.	mm	in.	mm	in.	mm
092S	285	129	324	147	233	106	283	128	248	112	289	131	303	137	330	150	46.25	1174	47.25	1200	27	686	285	711
102S	285	129	324	147	233	106	283	128	248	112	289	131	303	137	330	150	46.25	1174	47.25	1200	27	686	285	711
120S	283	128	322	146	228	103	277	126	274	124	320	145	341	155	372	169	43.25	1099	43.25	1099	26.75	679	27.75	705

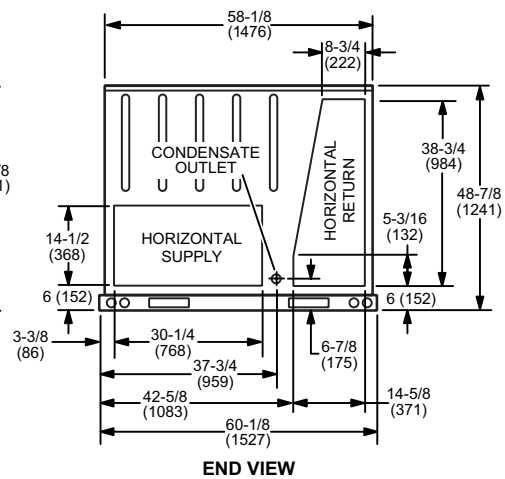
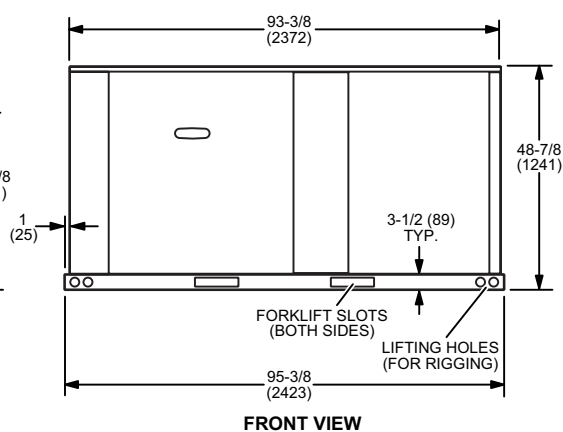
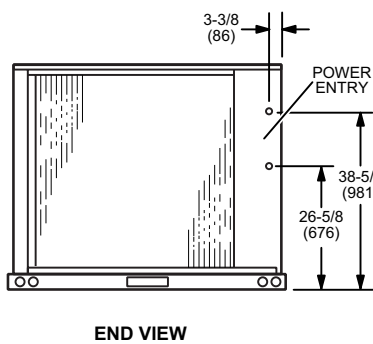
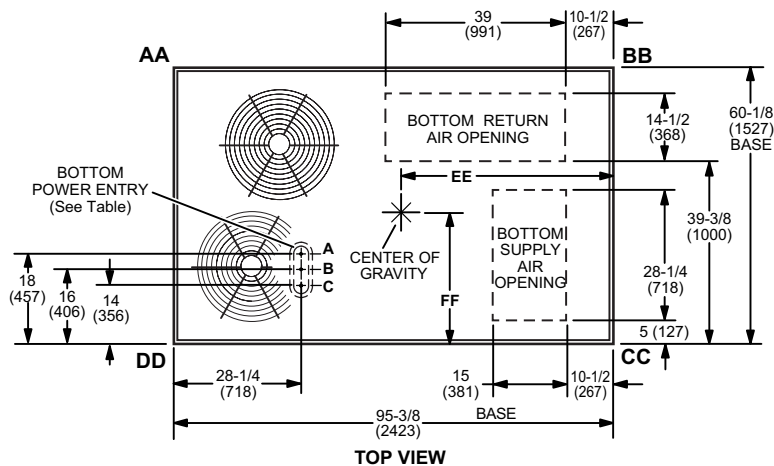
Base Unit - The unit with NO OPTIONS.

Max. Unit - The unit with ALL OPTIONS Installed. (Economizer, etc.)

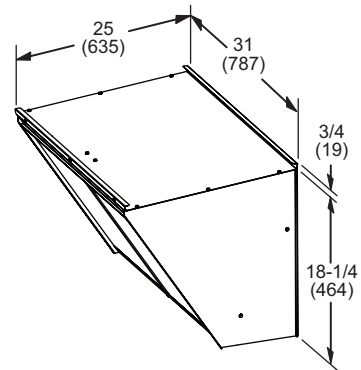
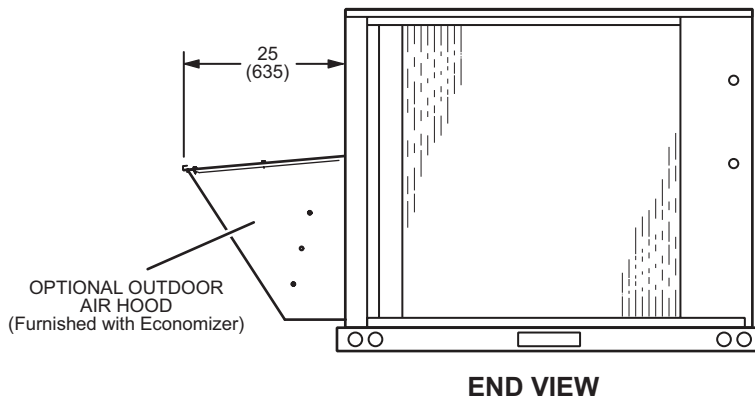
BOTTOM POWER ENTRY

Holes required for Optional Bottom Power Entry Kit

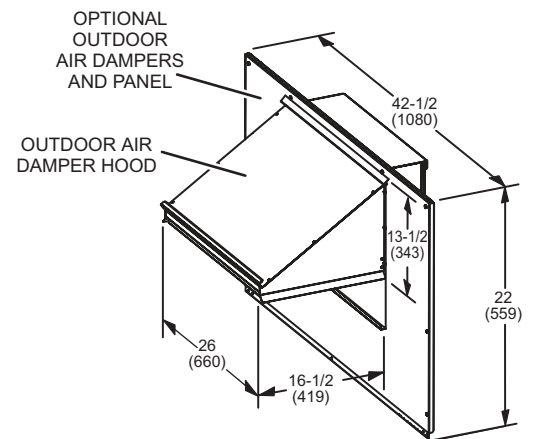
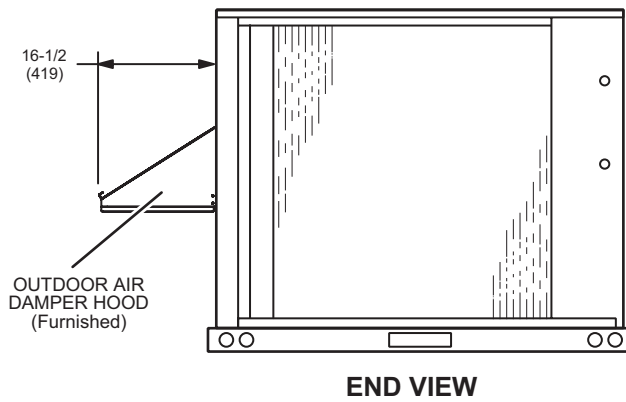
	Threaded Conduit Fittings (Provided in Kit)	Wire Use	Hole Diameter Required in Unit Base (Max.)
A	1 1/2	ACC	7/8 (23)
B	1/2	24V	7/8 (23)
C	1-1/4	POWER	1-3/4 (44)

¹ Field provided.

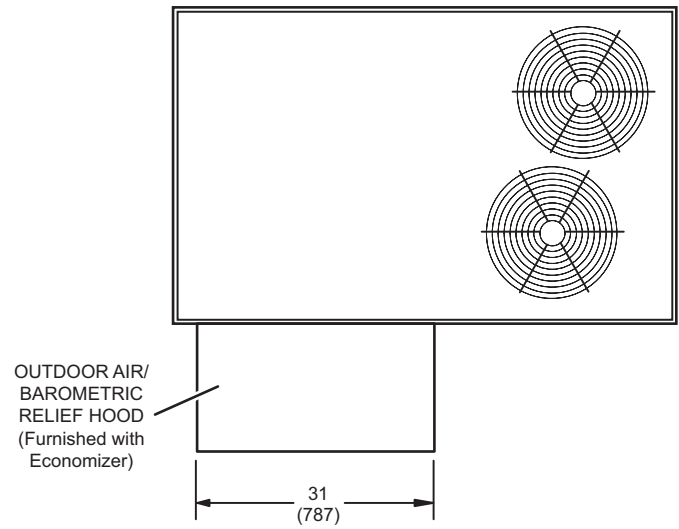
**OUTDOOR AIR HOOD DETAIL FOR OPTIONAL DOWNFLOW ECONOMIZER
(Downflow Applications)**



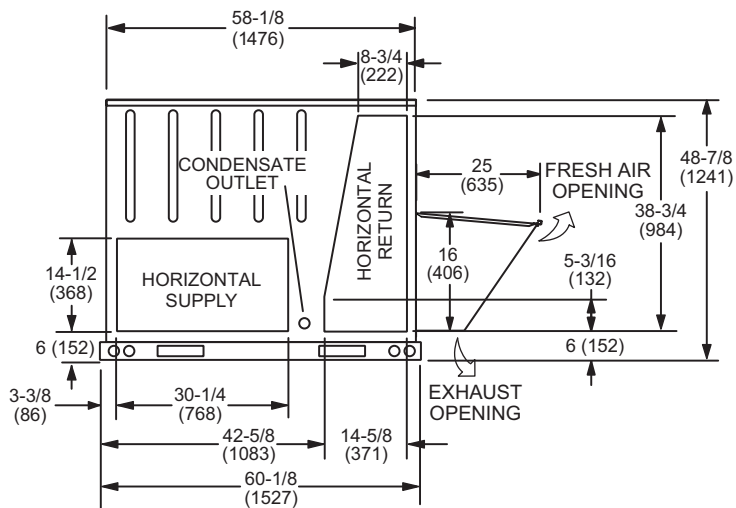
**OUTDOOR AIR DAMPER HOOD DETAIL FOR OPTIONAL MANUAL OR MOTORIZED OUTDOOR AIR DAMPERS
(Downflow or Horizontal Applications)**



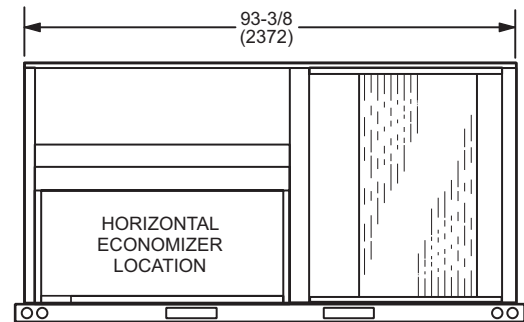
**OUTDOOR AIR HOOD DETAIL WITH OPTIONAL HORIZONTAL ECONOMIZER AND BAROMETRIC RELIEF DAMPERS
(Horizontal Applications)**



TOP VIEW



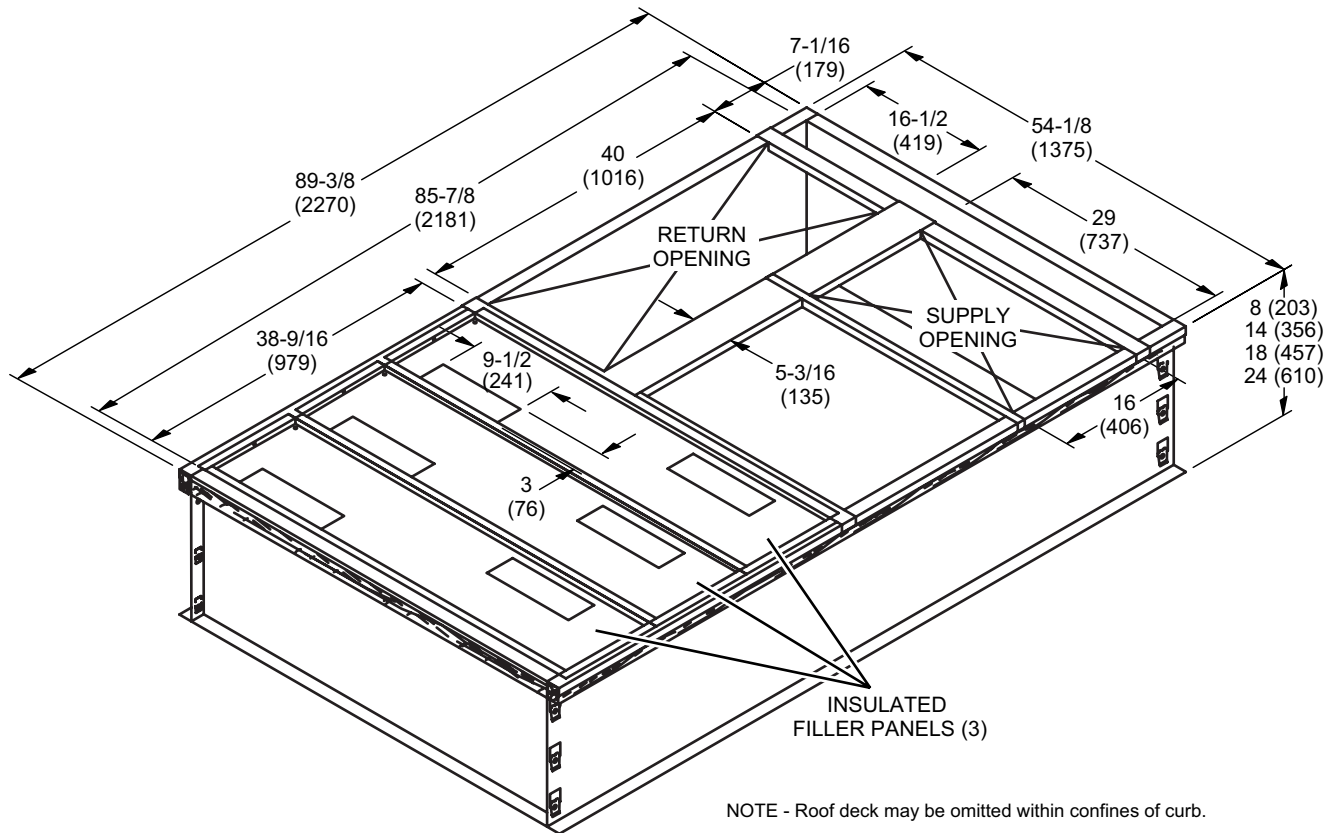
END VIEW



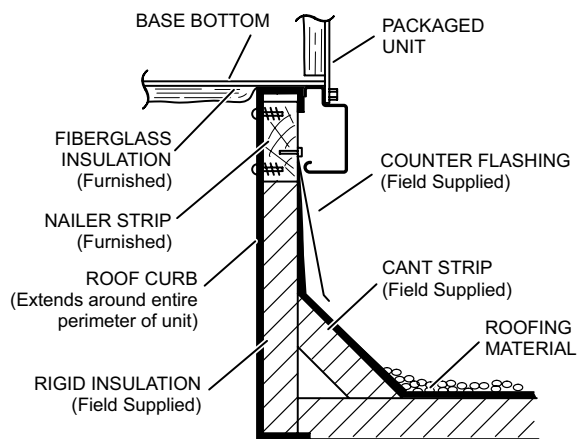
BACK VIEW

Note - Supply and Return Air Ducts must be supported.

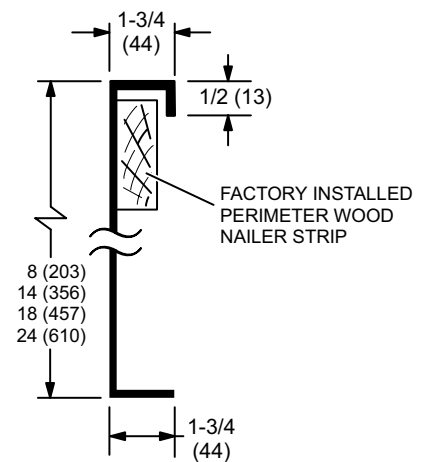
HYBRID CURBS - DOUBLE DUCT OPENING



TYPICAL FLASHING DETAIL FOR ROOF CURB

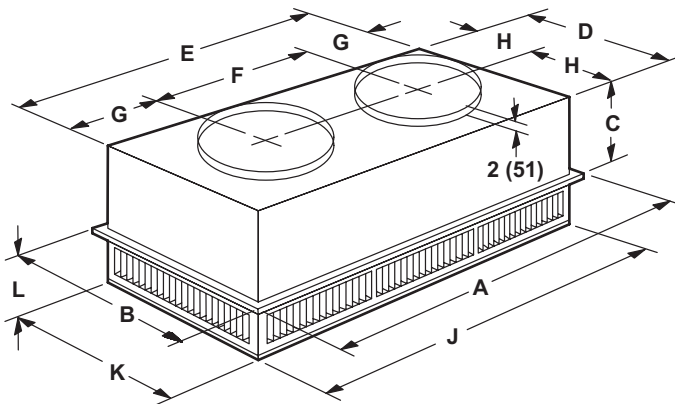


DETAIL ROOF CURB

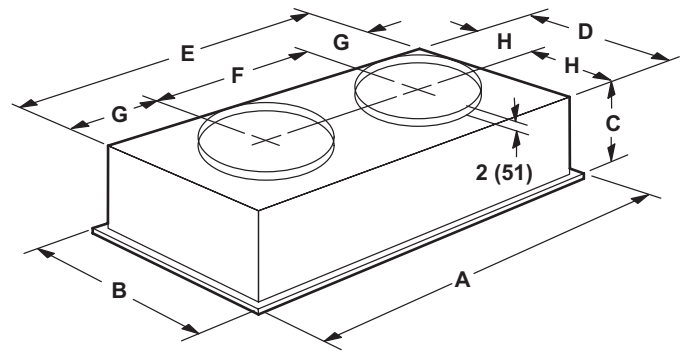


COMBINATION CEILING SUPPLY AND RETURN DIFFUSERS

STEP-DOWN CEILING DIFFUSER



FLUSH CEILING DIFFUSER

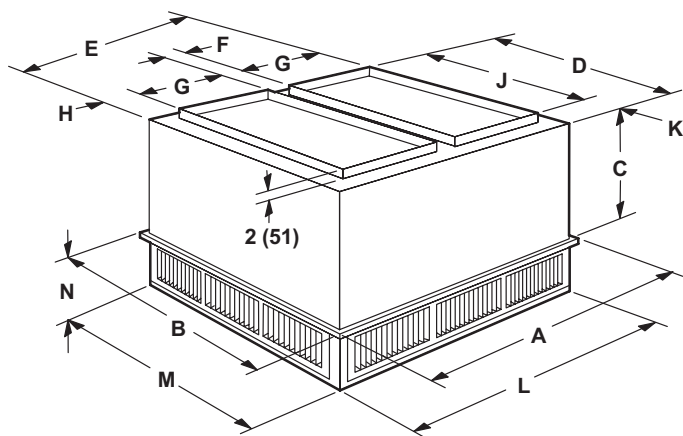


Model		RTD11-95S
A	in.	47-5/8
	mm	1159
B	in.	29-5/8
	mm	752
C	in.	14-3/8
	mm	365
D	in.	27-1/2
	mm	699
E	in.	45-1/2
	mm	1158
F	in.	22-1/2
	mm	572
G	in.	11-1/2
	mm	292
H	in.	13-3/4
	mm	349
J	in.	45-1/2
	mm	1156
K	in.	27-1/2
	mm	699
L	in.	8-1/8
	mm	206
Duct Size	in.	20 round
	mm	508 round

Model		FD11-95S
A	in.	47-5/8
	mm	1159
B	in.	29-5/8
	mm	752
C	in.	16-5/8
	mm	422
D	in.	27
	mm	686
E	in.	45
	mm	1143
F	in.	22-1/2
	mm	572
G	in.	11-1/4
	mm	286
H	in.	13-1/2
	mm	343
Duct Size	in.	20 round
	mm	508 round

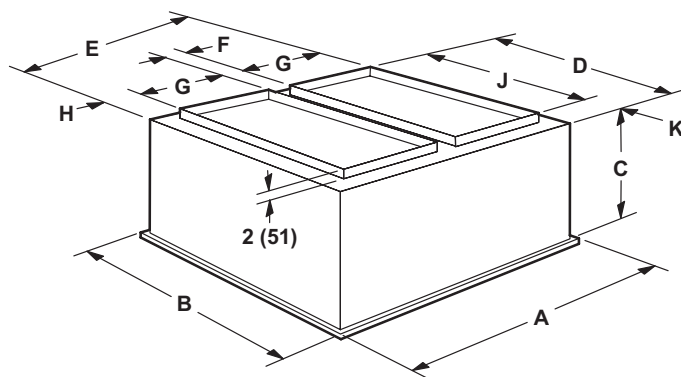
COMBINATION CEILING SUPPLY AND RETURN DIFFUSERS

STEP-DOWN CEILING DIFFUSER



Model		RTD11-135S
A	in.	47-5/8
	mm	1210
B	in.	35-5/8
	mm	905
C	in.	20-5/8
	mm	524
D	in.	33-1/2
	mm	851
E	in.	45-1/2
	mm	1156
F	in.	4-1/2
	mm	114
G	in.	18
	mm	457
H	in.	2-1/2
	mm	64
J	in.	28
	mm	711
K	in.	2-3/4
	mm	70
L	in.	45-1/2
	mm	1156
M	in.	33-1/2
	mm	851
N	in.	9-1/8
	mm	232
Duct Size	in.	18 x 28
	mm	457 x 711

FLUSH CEILING DIFFUSER



Model		FD11-135S
A	in.	47-5/8
	mm	1210
B	in.	35-5/8
	mm	905
C	in.	23-1/4
	mm	591
D	in.	33
	mm	838
E	in.	45
	mm	1143
F	in.	4-1/2
	mm	114
G	in.	18
	mm	457
H	in.	2-1/4
	mm	57
J	in.	28
	mm	711
K	in.	2-1/2
	mm	64
Duct Size	in.	18 x 28
	mm	457 x 711



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